

Ibn Tofail University

Rigid body mechanics

Problem Set V

Exercise 1:

1. State Koenig's theorem relating to the angular momentum of a perfect rigid body S , in motion relative to a reference frame T_1 .
2. Prove this theorem.

Correction

Exercise 2:

A system S consists of a homogeneous rod S_1 of mass m_1 and length l , in motion, in the plane $O_1X_1Y_1$, about the axis O_1Z of the frame $T_1(O_1, X_1, Y_1, Z_1)$, and a disk S_2 , of mass m_2 , radius R , and center C , in motion about the axis CZ_1 (fig 1). We denote by θ the angle that indicates the position of the rod with respect to T_1 and by φ the angle that indicates the position of the disk with respect to T_1 . $(\vec{i}, \vec{j}, \vec{k}_1)$ is the basis of the frame $T(O, X, Y, Z_1)$ attached to the rod.

1. Determine the position of the center of mass G of the system.
2. Calculate the kinetic elements of system S with respect to T_1 , at point O_1 .

Correction

Exercise 3:

A homogeneous rod of length l and mass m , rotates, in the vertical plane XOY , about the axis OZ of a reference frame $R(O, X, Y, Z)$ assumed to be Galilean. The position of the rod with respect to R is indicated by the angle θ (fig 2). We admit that the torsor of contact actions at O exerted on the rod reduces to the reaction $\vec{R}$.

1. By applying the angular momentum theorem to the rod, at O , find a first integral of motion, knowing that $\dot{\theta} = \omega_0$ when $\theta = 0$.
2. By applying the momentum theorem, determine the expression of the reaction.

Correction**Exercise 4:**

A homogeneous bar AB , of length $2l$ and mass m , rests on a point E located at a height h above the horizontal plane as shown in figure 3. Under the action of the force $\vec{F}_A$, point A moves on the axis OX with a constant velocity v_A . The contacts at points A and E are frictionless. The position of the bar, with respect to the terrestrial reference frame $R(O, X, Y, Z)$ assumed to be Galilean, is defined by the angle θ . The distance OH is constant.

1. Express $\dot{\theta}$ and $\ddot{\theta}$ as a function of θ and v_A .
2. Determine, in the basis of R and as a function of $\dot{\theta}$, the momentum of bar AB and its angular momentum at point A .
3. By applying the angular momentum theorem to bar AB at point A , determine the expression of the reaction $\vec{R}_E$ exerted on the bar at point E .

Correction**Exercise 5:**

A material point A of mass m is fixed on the perimeter of a hoop D , of center C , radius R , and mass M . This system S rolls without slipping on the axis OX of a terrestrial reference frame $R(O, X, Y, Z)$ assumed to be Galilean. The motion takes place in the plane XOY . We denote by θ the position of point A with respect to the hoop D , and I the geometric contact point (fig 4).

1. What is the number of degrees of freedom of the system?
2. Calculate the velocities of points I , C , and A with respect to R .
3. Find the momentum of S .
4. Determine the differential equation of motion, in θ , of S by applying the angular momentum theorem at point I .
5. What is the nature of the motion of S ?

Correction